# Coursework for
# 4G3 Computational Neuroscience
# Assignment 1

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assigned: February 13, 2026
**due: February 27, 2026, 4pm**
**submit through Moodle**
**don't forget to attach the correct cover page**

In this handout,

- text in blue corresponds to things you have to do / implement. To perform your simulations, you may use any programming language of your choice.
- text in orange corresponds to questions you need to answer in some form in your report.

Please write up your findings in a report to be submitted on Moodle in PDF format, and *include all your code in the Appendix*. Please do not forget to include the correct cover page (which depends on your course, see the cover page section on Moodle) and clearly include either your candidate number (all undergraduate students), or blind grading number (all postgraduate students). Do not use your name (or crsid) either in the form or in the name of the file you are submitting, as **your submission must remain anonymous**.

Your report should address all the questions raised in this handout, be structured around the sections herein, and **be a maximum of eight A4 pages** excluding any Appendix (minimum font size 11pt, minimum margins 1.5cm on each side, and you are encouraged to make your submission shorter than this maximum as long as you solve all exercises). You are very much encouraged to think of data/results visualisations to best support the exposition and interpretation of your results.

Please make sure to include a statement at the beginning of your report declaring whether you used generative AI tools (e.g. ChatGPT) for your simulations and / or writing your report. If you did, prepare an Appendix including all prompts you entered while using these tools in connection with this Assignment (including for literature search, explaining scientific concepts, help with code writing, help with writing / polishing the text or the figures of the report). As a word of caution, use these tools with great care: you will be responsible for any result you present and, if asked, you will need to be able to give evidence that you understand how exactly those results were obtained.

# I. Reinforcement learning

**[30 points]**

In this exercise you will study simple generalisations of the temporal-difference (TD) learning algorithm we discussed in lectures.

The form of the TD algorithm presented in the lectures builds up an estimate of the value for each separate state, thus leading to a "lookup table" representation. This is infeasible when the state space is too large, for two reasons. First, it might require us to store too many values. Second, and more fundamentally, typically several visits to the same state are needed to build a reliable estimate of its value, but we may never enter the exact same state twice (or enough times). This could be circumvented if knowledge about the value of one state could inform estimates for other states – i.e. if we had a representation that allowed *generalisation* across states.

A classical way to achieve generalisation, which has recently become very popular in cutting-edge machine learning applications, is based on the idea of *function approximation*. That is, instead of explicitly representing the (approximate) value function, $\hat{V}^\pi(s)$, for each state, $s$, and estimating these values directly, we use some parametric function to approximate the value function, $\hat{V}^\pi(s) = f(s; \boldsymbol{\theta})$, and iteratively estimate its parameters, $\boldsymbol{\theta}$ (we use boldface to denote vectors), using the same bootstrapping idea underlying lookup table-TD:

$$\theta_k \leftarrow \theta_k + \epsilon \left[ r_t + \gamma \hat{V}^\pi(s_{t+1}) - \hat{V}^\pi(s_t) \right] \frac{\partial f(s_t, \boldsymbol{\theta})}{\partial \theta_k} \tag{1}$$

where $t$ indexes time, $r$ is immediate reward, and $\gamma$ is the discounting factor – as in the lecture handouts.

1. Show that Equation 1 reduces to the lookup table-form of TD discussed during lectures with the appropriate choice of $f(s; \boldsymbol{\theta})$. What is the form of $f(s; \boldsymbol{\theta})$ in this case, and what are its parameters?

A special case of function approximation is *linear function approximation*, when the function approximator is written as a weighted sum of the output of "feature detectors", $\phi_k(s)$, each computing some simple and fixed function of $s$ (detecting some feature in it), such that the outputs of these feature detectors collectively represent $s$ in a format that is more conducive to learning (for example, when $s$ becomes high dimensional, as we will see below). The learnable parameters of a linear function approximator are the weighting factors, $\mathbf{w}$, controlling the relative contribution of each of these features to the value function estimate: $f(s, \mathbf{w}) = \sum_k w_k \, \phi_k(s)$.

2. What is the form of the update shown in Equation 1 in this special case?

In the lectures, the way we mapped the abstract notion of a "state" in TD (and, generally, in Markov decision processes) to a classical conditioning experiment, was that we assumed that each state corresponds to one time step in a trial of the experiment. However, the (often implicit) requirement for TD to work (and in fact, the assumption underlying essentially all reinforcement learning algorithms) is the so-called Markov property of the states on which it operates: that the future of a system should be independent of its past given its current state (and the currently executed action). This poses a problem for our mapping because in

a generic classical conditioning experiment what happens in future time steps may not only depend on whatever is happening in the current time step, but also what happened over some history of past time steps (which stimulus, if any, was presented, exactly how long ago, etc). Thus, our original naïve mapping from time steps to states is only appropriate as long as every trial goes through the exact same sequence of events, with a perfectly repeatable sequence of (conditioned) stimuli.

To handle the more general situation with potential stochasticity in stimuli, etc, we need to use a richer state variable, for example one which includes a record of the past history of stimuli as the state. The most straightforward such state variable is a stimulus-by-time step matrix, $\mathbf{S}$, containing the recent history of the stimulus, such that $S_{i\tau}$ is the extent to which stimulus $i$ was present $\tau$ time steps ago, with $0 \leq \tau \leq T_{\text{mem}}$, where $T_{\text{mem}}$ is the memory span of the agent. (Note that $\tau$ indexes time relative to the current "absolute" time, $t$, in the reverse direction: $s_t = \mathbf{S}$, such that $S_{i\tau} = y_i(t-\tau)$, where $y_i(t)$ is the presence of stimulus $i$ as a function of absolute time.) Note that $\mathbf{S}$ now spans a potentially very large state space which motivates the use of function approximation.

3. A *"tapped delay line" representation* uses feature detectors that are indexed by both stimulus and time such that the feature detector index introduced above is the tuple $k = \{i, \tau\}$, and $\phi_{i\tau}(\mathbf{S}) = S_{i\tau}$, i.e. each feature detector simply returns the level of one stimulus at a specified time ago. Using this representation, and the update rule derived for a linear function approximator in Exercise I.2, implement the TD algorithm. Simulate $N = 201$ trials of an experiment with (within-trial) time going from $t = 0$ to $25$ s (in steps of $\Delta t = 0.5$ s), a single stimulus presented at 10 s, such that its level is described by a "spike" $y_i(t) = g(t - 10\,\text{s})$, where $g(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$, and reward presented in a temporally smoother way around 20 s, such that its level varies as $r(t) = \frac{1}{2}\,\mathcal{G}(t;\, 20\,\text{s},\, 1\,\text{s})$ where $\mathcal{G}(x;\, \mu, \sigma) = e^{-\frac{(x-\mu)^2}{2\,\sigma^2}}$ is the Gaussian function with mean $\mu$, standard deviation $\sigma$, and maximal value 1. Use $\gamma = 1$ (i.e. no discounting), a learning rate $\epsilon = 0.2$ (for the given time step size), and a memory span $T_{\text{mem}} = 12$ s (so that $\mathbf{S}$ is a $1 \times 25$ matrix).

   (a) Plot the stimulus and the reward as a function of time.

   (b) For every 10th trial (starting from the first one, so 21 example trials in total), plot the following key variables as functions of time, $t$:

      i. the value, $\hat{V}(t)$,

      ii. the temporal difference of the value $\Delta\hat{V}(t) = \gamma\,\hat{V}(t) - \hat{V}(t - \Delta t)$, and

      iii. the temporal difference learning error $\delta(t) = r(t - \Delta t) + \Delta\hat{V}(t)$.

      Preferably, use the format seen in lecture slides: three panels underneath each other for the three variables, sharing the same x- (time) axis. In each panel, you can use lines with different colours corresponding to the 21 example trials that need to be plotted.

   (c) Comment on your results. Why do the time courses of the variables look like they do, and why do these time courses change across learning in these particular ways?

   (d) The general observation in actual data about dopamine cell activation discussed during lectures is that before (or at the beginning of) learning, dopamine cells are active at the time of the reward, while after (or towards the end of) learning they are active at the time of the stimulus. However, this "shift" happens in a way that no

dopamine activation is usually seen between these two times: instead, the "bump" of activity at the time of the reward gradually disappears while a bump at the time of the stimulus gradually develops. What variable in the TD model is supposed to correspond to dopamine activity? Does its behaviour in your model agree with these observations?

4. Another memory-based representation often used for (linear) function approximation is a *"boxcar" representation* in which feature-detectors are described as $\phi^{\square}_{i\tau}(\mathbf{S}) = \sum_{u=0}^{\tau} S_{iu}$, i.e. they detect the total presence of a stimulus over some past history. Using this representation, and a learning rate $\epsilon = 0.01$, simulate the same experiment (with all other parameters being the same) as above.

   (a) Plot the same three variables (in the same trials and in the same format) as before.

   (b) Comment on your results. Did anything change in the first and last trial from the previous results? Did anything change in the intervening trials, during the course of learning, compared with the previous results? If there were changes, explain why they occurred.

   (c) How do these results relate to the experimental observation described above about the "shift" of the dopamine signal over the course of learning?

   (d) Why did we need to change the learning rate? What would have happened if we had left it unchanged?

5. An often used classical conditioning paradigm uses *partial reinforcement* such that only some (randomly chosen) fraction, $p$, of trials ends with the reward presented, in the rest of the trials the reward is omitted. Simulate $N = 1000$ trials with $p = 0.5$, using the same model as above (with boxcar representation).

   (a) For the last 100 trials, plot the average of each of the same three variables as above, separately for

      i. rewarded trials,

      ii. unrewarded trials, and

      iii. for all trials.

   (b) Which variables show or don't show a difference between these three kinds of averages? Explain your findings.

   (c) Which of these variables is represented by the activity of dopamine neurons according to the TD theory of dopamine responses? What are the key features of the dopamine response that are well matched by this variable?

   (d) There have also been suggestions that the activity of dopamine neurons can be interpreted as representing surprise or salience. Explain with reasons if you think this is a correct interpretation of dopamine, with special regard to dopamine responses in rewarded vs. unrewarded trials.

   (e) Dopamine neurons have a limited and *asymmetric dynamic range*: they can only signal negative quantities (below their baseline) to a limited extent, and they also saturate at high firing rates so they can also only signal large positive quantities to a limited extent. Thus, a simple model of dopamine activity is:

$$\mathrm{DA}(x) = \begin{cases} x/\alpha & \text{if } x < 0 \\ x & \text{if } 0 \leq x < x^* \\ x^* + (x - x^*)/\beta & \text{if } x^* \leq x \end{cases}$$

where $x$ is the variable that dopamine activity encodes, and $\alpha = \beta = 6$ and $x^* = 0.27$. Plot the time course of the dopamine signal averaged across all the last 100 trials. How and why is it different from the average time course of the variable it encodes?

6. Simulate a set of similar partial reinforcement experiments, but each using a different reward probability: $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$.

   (a) Plot the average dopamine time course (in the last 100 trials, as above) for each of these experiments in a single figure (colour coding the corresponding value of $p$). What do you notice?

   (b) As a function of $p$, plot the (average) level of dopamine at its peak around the time of the stimulus, and also at the time of the reward. Describe in words how the level of dopamine depends on $p$ at these two times, and explain these dependencies.

   (c) There have been suggestions that dopamine might signal uncertainty, instead of a prediction error. What pattern in the dopamine signal you obtained might suggest this to be the case? Do you have an alternative explanation for this pattern?

7. Let us now consider a more complex visual stimulus, a time-varying $d \times d$ image (i.e. a movie). We can consider each pixel acting as an elementary stimulus, such that $y_{mn}(t)$ encodes the value of pixel $(m, n)$ at time $t$. The state variable storing stimulus history in this case becomes a tensor, $\mathbf{S} \in \mathbb{R}^{d \times d \times T_{\text{mem}}}$: $s_t = \mathbf{S}$, such that $S_{mn\tau} = y_{mn}(t - \tau)$, and the basic tapped delay line representation would use the simple feature detectors $\phi_{mn\tau}(\mathbf{S}) = S_{mn\tau}$.

   (a) Derive the boxcar representation for the visual stimulus. Let $\phi^{\square}_{mn\tau}(\mathbf{S})$ denote features of this representation.

   (b) Let us now consider a slightly more sophisticated representation that instead uses feature detectors $\phi^*_{i\tau}(\mathbf{S}; \mathbf{v}) = \sum_{u=0}^{\tau} h\left(\sum_{m,n=1}^{d} v^{(i)}_{mn} S_{mnu}\right)$, where $h(.)$ is some activation function (for example, the logistic sigmoid: $h(x) = (1 + e^{-x})^{-1}$), and the outputs of these feature detectors are still combined linearly: $f(\mathbf{S}, \boldsymbol{\theta}) = \sum_{i\tau} w_{i\tau}\, \phi^*_{i\tau}(\mathbf{S}; \mathbf{v})$, where the parameters of the representation are now $\theta = \{\mathbf{w}, \mathbf{v}\}$.

      i. Provide an intuitive explanation of the computation that each feature detector $\phi^*_{i\tau}(\mathbf{S}; \mathbf{v})$ performs, and in particular explain the meaning of its parameters $\{v^{(i)}_{mn}\}^{d}_{m,n=1}$ in the context of this computation.

      ii. Derive the update rule (corresponding to Equation 1) for the parameters of this representation for a general activation function, $h(\cdot)$. To simplify your expressions, use $h'$ to denote the derivative of $h(\cdot)$, and $a_{iu} = \sum_{m,n=1}^{d} v^{(i)}_{mn} S_{mnu}$ to denote the "feature pre-activations". Explain how this update rule is similar to or different from the one you derived in Exercise I.2.

      iii. Repeat your derivations for an identity activation function $h(x) = x$. Express your update rule in terms of the boxcar features $\phi^{\square}_{mn\tau}(\mathbf{S})$. Discuss any advantages or disadvantages that $\phi^*$ with $h(x) = x$ may have compared to $\phi^{\square}$.

      iv. Henschke et al. (2020)[1] showed experimentally that reward association enhances stimulus-specific representations in primary visual cortex (V1). In their experiment, responses of neurons in V1 to four grating stimuli were measured in two

[1] Henschke JU, Dylda E, Katsanevaki D, Dupuy N, Currie SP, Amvrosiadis T, Pakan JM, Rochefort NL. (2020) Reward association enhances stimulus-specific representations in primary visual cortex. *Current Biology* 30:1866–80.

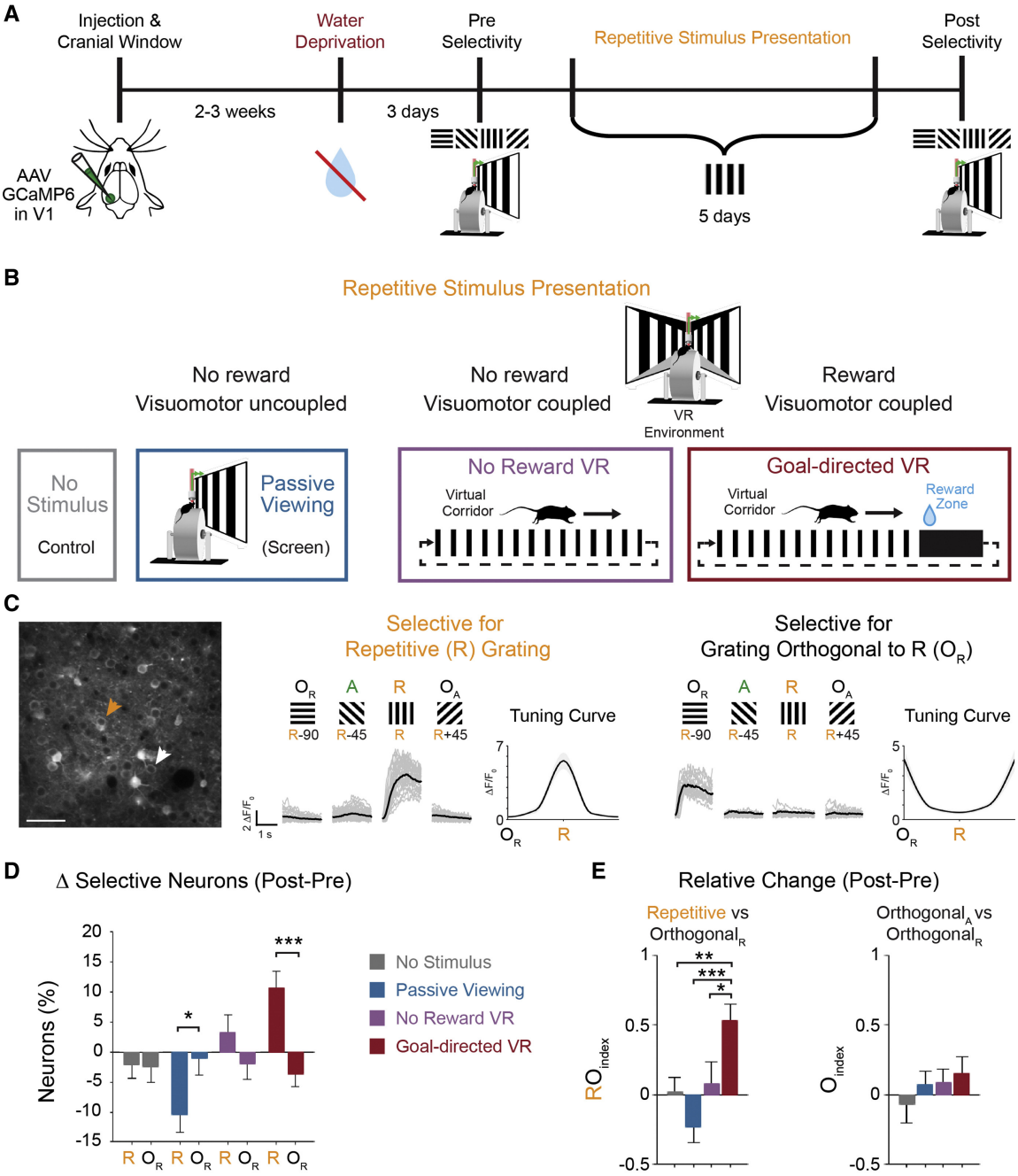


Figure 1. Reproduced from Henschke et al. (2020)[1].
*See caption on next page.*

Figure 1. *See figure on previous page.*
**(A)** Experimental timeline. The activity of neurons in V1 was recorded in head-fixed mice able to freely run on a cylindrical treadmill, and responses to 4 oriented gratings were assessed before (pre) and after (post) an intervening 5 consecutive days of repetitive stimulus presentation of one oriented grating.
**(B)** During the repetitive visual stimulus over the 5 days, animals were divided into experimental groups: *no stimulus presentation*; *passive viewing* condition with one grating at a constant temporal frequency (schematic of setup with single screen); and same grating in a virtual reality (VR) environment where visuomotor feedback is coupled and with either no reward (*no reward VR*) or a learning task with water reward (*goal-directed VR*).
**(C)** Example view of recorded neurons (left) with arrows indicating two example neurons that displayed orientation-selective response for either the stimulus that was repetitively presented (R) (middle, orange) or for the grating orthogonal to the repetitive grating ($O_R$) (right, gray). Responses of the example neurons to the 4 oriented gratings for each trial are shown in grey with average response across trials in black (A, angled; $O_A$ orthogonal to A). Tuning curves show the peak response at the preferred orientation. Scale bar: 50 mm.
**(D)** Change in the proportion of selective neurons from pre to post day for the R and $O_R$ gratings (*no stimulus*: p=0.918; *passive viewing*: p=0.030; *no reward VR*: p=0.199; *goal-directed VR*: p<0.001; Student's t test).
**(E)** Relative change from pre to post day between the R and $O_R$ gratings ($RO_{index}$; left; p=0.001; one-way ANOVA with LSD test) and other control orientations ($O_A$ and $O_R$; $O_{index}$; right; p=0.633; one-way ANOVA).
All panels: *p<0.05, **p<0.01, ***p<0.001; n=9 mice for each group; error bars: SEM

sessions (Figure 1A). Over a 5-days interval between the two measurement sessions, animals were either (i) passively exposed to a specific grating (vertical orientation), (ii) were engaged in a motor task during which this specific grating was presented or (iii) were additionally rewarded at the end of each trial (Figure 1B). In group (iii), but not the other groups, an elevated selectivity to the vertical orientation was observed (Figure 1C-E). Explain how this can be modeled in the TD framework based on your previous derivations. In your answer, specify which component of the model would correspond the V1 neural responses, and why the model would then explain how neural responses change over the experiment. **[150 words max.]**

8. In another study, Engelhard et al. (2019)[2] showed that dopamine neurons are universally responsive to rewards but highly specialised in their response to a variety of "behavioural variables" (Figure 2). In the context of this exercise, we assume these behavioural variables to be informative of future rewards, and so they can be considered a part of the "stimulus" in the TD model.

   (a) Explain with reasons if standard TD theory can account for such feature-specific dopamine responses.

   (b) Consider a linear function approximation scheme (as in Exercise I.2), in which we compute each feature-specific component of the value function explicitly, $\hat{V}_k^\pi(s) =$

[2]Engelhard B, Finkelstein J, Cox J, Fleming W, Jang HJ, Ornelas S, Koay SA, Thiberge SY, Daw ND, Tank DW, Witten IB. (2019) Specialized coding of sensory, motor and cognitive variables in VTA dopamine neurons. *Nature* 570:509–13.

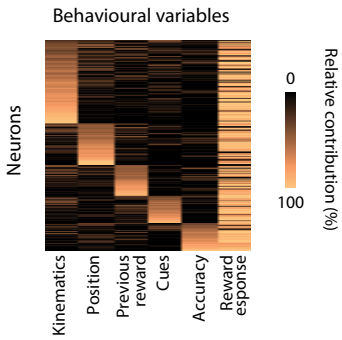


Figure 2. Reproduced from Engelhard et al. (2019)[2]. Relative contribution of each behavioural variable (columns) to the explained variance of the activity of each neuron (rows). For illustrative purposes, neurons were ordered by their selectivity for behaviour variables.

$w_k\,\phi_k(s)$, before we sum them up in the total value function: $\hat{V}^\pi(s) = \sum_k \hat{V}_k^\pi(s)$. Suggest a meaningful mathematical definition for a feature-specific TD prediction error, $\delta_k$, such that it only refers to the corresponding feature-specific component of the value function, $\hat{V}_k^\pi(\cdot)$, and show how these feature-specific TD prediction errors can be used to compute updates to the feature-specific parameters, $w_k$, in a way that these updates are equivalent to those of the standard TD model. Explain with reasons if this model can account for the feature-specific dopamine responses found by Engelhard et al. (2019)[2], as well as if it is "backward compatible" with the standard TD model in that it reproduces all the main effects regarding dopamine responses discussed during lectures and in this assignment. You are welcome to support your answer with simulations.